**SECTION–I(i) : (Maximum Marks : 24)**

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +3 If only the bubble corresponding to the correct option is darkened.
*Zero Marks* : 0 If none of the bubbles is darkened.
*Negative Marks* : –1 In all other cases

**1.** Metals which are not dissolved in conc. $HNO_3$ due to formation of passive film of oxide.
(A) Hg, Cu (B) Ag, Zn (C) Au, Pt (D) Al, Cr

**2.** Which of the following metal cation gives green colour in the borax bead test in oxidizing flame in cold condition.
(A) Cu (B) Co (C) Cr (D) Ni

**3.** Which of the following salt produce coloured gas with dil.$H_2SO_4$ ?
(A) NaBr (B) $NaNO_2$ (C) $NaNO_3$ (D) NaI

**4.** Oxidation number of silver is **NOT** changed when –
(A) $Ag_2SO_{3\,(Aq)}$ is boiled
(B) $Ag_2CO_3$ crystal is strongly heated
(C) $AgNO_3$ reacts with aqueous $[Co(NH_3)_6]Cl_3$ solution
(D) $AgNO_{3\,(Aq.)}$ solution reacts with Cu-metal

**5.** Which of the following cation produce white coloured ppt. with dil. HCl :-
(A) $Hg_2^{2+}$ (B) $Pb^{2+}$ (C) $Ag^+$ (D) All of these

**6.** If in sodium salt 'A' anion have oxidation state of central atom is +3. Salt 'A' react with conc. $H_2SO_4$ and produce two gaseous product (no one is $SO_2$). Select the **CORRECT** alternative about salt 'A' :-

(A) $Salt_{(aq.)}$ 'A' + $KMnO_4/H^+ \xrightarrow[\Delta]{60^\circ}$ brown ppt.

(B) $Salt_{(aq.)}$ 'A' + $\underset{\text{neutral sol.}}{FeCl_3}$ → deep red colour sol.

(C) $Salt_{(aq.)}$ 'A' + dil.$H_2SO_4$ → no observable change

(D) Salt 'A' contains $CH_3COO^-$ anion

**7.** Select the **INCORRECT** match :-

| Test of cations | Group Reagent |
|---|---|
| (A) $Cu^{+2}$, $Cd^{+2}$, $Hg^{+2}$ | $H_2S$ with dil. HCl |
| (B) $Fe^{+3}$, $Cr^{+3}$, $A\ell^{+3}$ | $NH_4OH + NH_4Cl$ |
| (C) $Ba^{+2}$, $Sr^{+2}$, $Ca^{+2}$ | $(NH_4)_2CO_3 + NH_4Cl$ |
| (D) $Mn^{+2}$, $Zn^{+2}$, $Co^{+2}$ | $H_2S$ in acidic medium |

**8.** (X) + Zn powder $\xrightarrow[\Delta]{OH^\ominus}$ $'Y'_{(g)}$ $\xrightarrow{\text{dil. HCl}}$ white fumes

salt
↓ sulphamic acid
$Z_{(g)}$

Identify 'Y' & 'Z' respectively
(A) $SO_2$ , $H_2S$ (B) $N_2O$, $NH_3$ (C) $NH_3$, $N_2$ (D) $N_2$, $N_2O$



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## SECTION-I(ii) : (Maximum Marks: 12)

- This section contains **THREE** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

| | | |
|---|---|---|
| *Full Marks* | : +4 | If only (all) the correct option(s) is (are) chosen. |
| *Partial Marks* | : +3 | If all the four options are correct but ONLY three options are chosen. |
| *Partial Marks* | : +2 | If three or more options are correct but ONLY two options are chosen, both of which are correct options. |
| *Partial Marks* | : +1 | If two or more options are correct but ONLY one option is chosen and it is a correct option. |
| *Zero Marks* | : 0 | If none of the options is chosen (i.e. the question is unanswered). |
| *Negative Marks* | : –2 | In all other cases. |

**9.** Which of the following compound is/are produced during heating borax ($Na_2B_4O_7 \cdot 10H_2O$)
(A) $Na_2B_4O_7$(Anhydrous) (B) $NaBO_3$
(C) $NaBO_2$ (D) $B_2O_3$

**10.** $Na_2S_2O_3$(solution) $\xrightarrow{'X'/H^+}$ coloured solution
Where 'X' is/are -
(A) $CuSO_4$(aq.) (B) $FeCl_3$(aq.) (C) $Cl_2$ water (D) $Cr_2O_7^{-2}$

**11.** Aqueous solution containing $Al^{+3}$ , $Mg^{+2}$, $Fe^{+3}$, $Ni^{+2}$ is treated with $NH_4Cl$ & $NH_4OH$. Which ion(s) is/are precipitated
(A) $Al^{+3}$ (B) $Mg^{+2}$ (C) $Fe^{+3}$ (D) $Ni^{+2}$

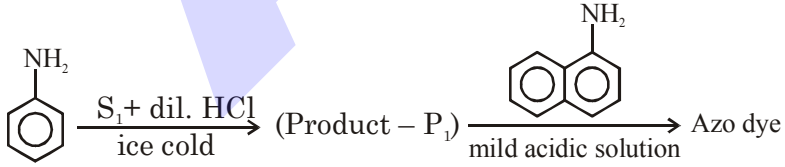
## SECTION–I(iii) : (Maximum Marks : 12)

- This section contains **ONE** paragraph.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

| | | |
|---|---|---|
| *Full Marks* | : +3 | If only the bubble corresponding to the correct answer is darkened. |
| *Zero Marks* | : 0 | In all other cases. |

### Paragraph for Q.12 to Q.13

Consider two potassium salt ($S_1$) and ($S_2$) which gives following observation




$CuSO_{4(aq.)} \xrightarrow{Salt - S_2}$ (Product – $P_2$)↓ + $K_2SO_4$ + poisonous gas (G)

(Product – $P_1$) + (Product –$P_2$) ⟶ Show Sandmeyer's reaction

**12.** Salt–$S_2$ is
(A) KCN (B) KBr (C) $KNO_2$ (D) NaCN

**13.** Select **CORRECT** for anion in salt $S_1$
(A) can react with $KMnO_4/H^+$ (B) react with $KI/H^+$
(C) gives $NH_3$ with KOH + Al (D) All are correct

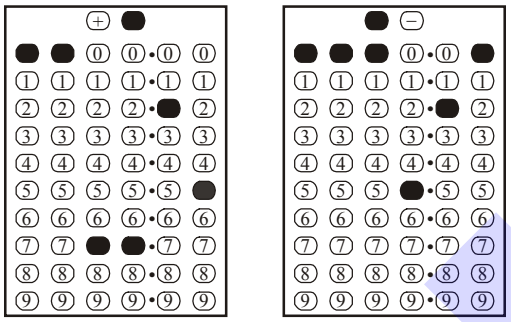


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## SECTION-II : (Maximum Marks: 12)

- This section contains **FOUR** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

**For Example :** If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 If ONLY the correct numerical value is entered as answer.
*Zero Marks* : 0 In all other cases.

**1.** Find the total number of oxidising agents which can oxidise $SO_2$ to $SO_4^{-2}$ in acidic medium.
$Fe^{+2}$ , $Cr_2O_7^{-2}$ , $MnO_4^-$ , $Br_2$ water , $H_2S$ , $KIO_3$ , $Cl_2$ water , $H_2O_2$

**2.** How many of the following salt on heating in bunsen flame gives characteristic green flame:-
$BaCl_2$ , $CaCl_2$ , LiCl , NaCl , $MgCl_2$

**3.** $Na_2CrO_{4(Aq.)} \xrightarrow{\text{conc. } H_2SO_4}$ product
What is the difference between oxidation number of Cr in product and reactant :-

**4.** Find the number of yellow colour sulphide(s) in the following :-
CdS, $Sb_2S_3$, $As_2S_3$, $As_2S_5$, $SnS_2$, ZnS, PbS

## SECTION–IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II.**
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II.**
- One or more entries in **Column-I** may match with one or more entries in **Column-II.**
- The ORS contains a 4 × 5 matrix whose layout will be similar to the one shown below :

(A) (P) (Q) (R) (S) (T)
(B) (P) (Q) (R) (S) (T)
(C) (P) (Q) (R) (S) (T)
(D) (P) (Q) (R) (S) (T)

- For each entry in **column-I,** darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**
*Full Marks* : +2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened
*Zero Marks* : 0 In all other cases



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**1.**

| Column I | Column II |
|---|---|
| (A) $S_2O_3^{2-}$ | (P) give fumes/gas with dil $H_2SO_4$ |
| (B) $NO_3^{\ominus}$ | (Q) give violet coloured ppt. with $[Ni(en)_3](NO_3)_2$ |
| (C) $NO_2^{\ominus}$ | (R) give white ppt with $BaCl_2$ solution |
| (D) $S^{2-}$ | (S) identified by brown ring test |
| | (T) identified by methylene blue dye test |

**2.** Match the folloiwng column :

| **Column - I (Test of cation/anion)** | **Column - II (Observation)** |
|---|---|
| (A) $NO_2^-$ + dil. $H_2SO_4$ + $FeSO_4$(Aq.) | (P) Purple colour solution |
| (B) $S^{-2}$ + $Na_2[Fe(CN)_5NO]$ | (Q) Brown ring formation |
| (C) Flame test of $Ba^{2+}$ | (R) Colourless solution |
| (D) $Pb^{+2}$ + NaOH (excess) | (S) Apple green colour flame |
| | (T) Complex compound is formed in product side |

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